

Embryonic, Organ and Other Tissue Specific Stem Cells

Clinical analysis of fecal microbiota transplantation for refractory acute gastrointestinal graft-versus-host disease after allogeneic hematopoietic stem cell transplantation: a five-year retrospective study



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ABSTRACT

Background aims: Refractory acute gastrointestinal graft-versus-host disease (GI-aGVHD) is a severe complication following allogeneic hematopoietic stem cell transplantation (allo-HSCT), which is associated with significantly increased mortality. Fecal microbiota transplantation (FMT) has emerged as a promising therapeutic strategy for this condition, and its efficacy and safety have garnered substantial attention in clinical research.

Objective: To explore the clinical efficacy and safety of FMT for refractory acute GI-aGVHD after allo-HSCT.

Methods: Data from 22 patients with refractory GI-aGVHD who received FMT and 20 patients who did not undergo FMT, all diagnosed with refractory GI-aGVHD after allo-HSCT and treated at Henan Cancer Hospital between December 2017 and March 2023, were retrospectively analyzed to evaluate short-term efficacy, safety, and long-term survival outcomes.

Results: FMT significantly reduced the median diarrhea volume (1300 versus 450 mL, $P = 0.017$) and frequency (10 versus 5.5 times/d, $P = 0.049$) in the FMT group, whereas the control group showed no significant improvement in either parameter (diarrhea volume: 1350 versus 900 mL, $P > 0.05$; frequency: 10 versus 9 times/d, $P > 0.05$). The FMT group exhibited significantly higher complete response (CR) (31.8% versus 5%, $P = 0.047$) and overall response rate (ORR) (59.1% versus 25%, $P = 0.033$) compared with the control group. Subgroup analyses showed that patients with isolated intestinal involvement (ORR = 100%) and those without bloody stool before FMT (ORR = 78.6%) had superior responses to FMT. The FMT group achieved a median progression-free survival (PFS) of 30 (10–2113) days and a median overall survival (OS) of 150 (51–2130) days, with the 100-day, 1-year and 3-year PFS rates of 40.9%, 28.1%, and 28.1%, respectively, and the 100-day, 1-year and 3-year OS rates of 59.1%, 50.0% and 34.3%. In contrast, the control group had notably poorer survival outcomes: median PFS was 23 (7–1820) days, median OS was 80 (20–1901) days, while the 100-day, 1-year, and 3-year PFS rates remained 5% across all time points, and the 100-day, 1-year, and 3-year OS rates were 20.0%, 10.0%, and 10%.

Conclusion: FMT is effective and safe as a clinical treatment for refractory GI-aGVHD after allo-HSCT, and the effectiveness of FMT can be used as a prognostic indicator for long-term survival in refractory GI-aGVHD.

Key Words: allogeneic, fecal microbiota transplantation, graft-versus-host disease, hematopoietic stem cell transplantation.

Abbreviations: allo-HSCT, allogeneic hematopoietic stem cell transplantation; GI, gastrointestinal; GVHD, graft-versus-host disease; FMT, fecal microbiota transplantation; CR, complete remission; PR, partial remission; SD, stable disease; PD, progressive disease; ORR, overall response rate; PFS, progression-free survival; OS, overall survival; AML, acute myeloid leukemia; ALL, acute lymphoblastic leukemia; SAA, severe aplastic anemia; MDS, myelodysplastic syndromes; PTCL, peripheral T-cell lymphoma; CLL, chronic lymphocytic leukemia; HPD, haploidentical donors (HPD); MSD, matching-sibling donors; URD, unrelated donors; PBSC, peripheral blood stem cells; BM, bone marrow; UCB, umbilical cord blood; DLI, donor lymphocyte infusion

Introduction

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is an irreplaceable treatment for a variety of hematological diseases. Acute gastrointestinal graft-versus-host disease (GI-aGVHD) is a common complication of allo-HSCT. Grade III/IV GI-aGVHD seriously affects the prognosis and quality of life of the patients [1,2]. At present, the first-

line treatment for GI-aGVHD is mainly glucocorticoids, but nearly half of the patients are resistant to them, and the survival rate of patients with steroid-refractory GI-aGVHD (SR-GI-aGVHD) is only 5–30% [3–5].

Previous studies have shown that the gut microbiota plays an important role in the occurrence and development of aGVHD after allo-HSCT, and the loss of microbial diversity can promote the development of aGVHD [6,7]. Therefore, regulating the homeostasis of gut flora and maintaining the dominance of beneficial bacteria is a viable treatment for GI-aGVHD. Fecal microbiota transplantation (FMT) can rapidly restore the gut microbiota of the receptor, and increase regulatory T cells and short-chain fatty acids to repair the intestinal mucosal barrier, which resolves inflammatory responses and recalibrates the immune system, contributing to the repair of the intestinal mucosa [8,9]. Some small sample studies have also shown good efficacy of FMT in the treatment of refractory aGVHD [10,11].

Patients and Methods

Patients

From December 2017 to March 2023, the medical records of 22 patients with refractory GI-aGVHD who underwent allo-HSCT and received FMT, along with 20 matched patients without FMT, were retrospectively analyzed at the Affiliated Cancer Hospital of Zhengzhou University. The inclusion criteria for fecal bacteria transplantation need to meet the following criteria at the same time: (1) Gastrointestinal acute graft-versus-host disease was diagnosed after allogeneic hematopoietic stem cell transplantation; (2) Patients who have failed first-line treatment with standard glucocorticoids and at least one second-line immunosuppressive therapy (morbidity, CD25 monoclonal antibody, budesonide, ruxolitinib, etc.). According to the 2018 European Bone Marrow Transplantation Society-NIH-Centre for International Bone Marrow Transplantation Research (EBMT-NIT-CIBMTR) criteria [12], defined glucocorticoid resistance as disease progression assessed by efficacy within 3–5 days of first-line glucocorticoid therapy or non-response assessed by efficacy within 5–7 days of therapy or failure to achieve complete remission by 28 days of immunosuppressive therapy including glucocorticoids. Glucocorticoid dependence was defined as the inability to reduce the dose of first-line glucocorticoids or the reactivation of GVHD during the reduction process. Glucocorticoid

resistance and glucocorticoid dependence were collectively referred to as glucocorticoid treatment failure. Exclusion criteria: (1) Persistent, uncontrolled infection (i.e., bacteraemia, haemodynamic instability, persistent fever or within 3 days of antibiotic treatment after bacteraemia); (2) Persistent intestinal disease other than GI-aGVHD; (3) Gastrointestinal perforation; (4) Possible clinical contraindications identified by the investigator. All patients signed an informed consent form.

FMT procedures

The fecal materials were handled in sealed, fully automatic machines (GenFMTer, Nanjing, China), and a Good Manufacturing Practice (GMP)-level laboratory room. Donors aged between 6 and 24 years were selected from the Chinese Fecal Bacteria Bank with strict screening according to the guidelines [13]. The fecal bacteria were injected into the intestinal tract through a nasojejunal tube, and the frequency of infusion was determined according to the needs of the disease. FMT was initiated upon diagnosis of refractory GI-aGVHD. Infusions were administered every 3 days, and the treatment strategy was adjusted based on efficacy evaluations as follows: infusions were discontinued in cases of CR or PD; continued until best response was achieved in patients with PR; and discontinued if SD was confirmed after three cycles of infusion. A detailed decision algorithm is illustrated in the decision tree (Figure 1). Avoid antibiotics 12–48 h before fecal bacteria transplantation, and avoid broad-spectrum antibiotics if it is difficult to completely stop antibiotics. Bowel preparation: intravenous proton pump inhibitors 1 h before treatment and intramuscular metoclopramide 10 mg 30 min before treatment.

Evaluation of efficacy and safety

The short-term efficacy indicators were the frequency and amount of diarrhea 14 days and the clinical remission 21 days after diagnosing refractory GI-aGVHD. The response rate was obtained by comparing the GI-aGVHD grade and overall grade with the initial situation. Complete remission (CR) was defined as complete disappearance of GI-aGVHD manifestations; partial remission (PR) was defined as improvement in GI-aGVHD (at least one grade lower) without achieving CR; stable disease (SD) was defined as no improvement in the severity of GI-aGVHD nor worsening of intestinal aGVHD or

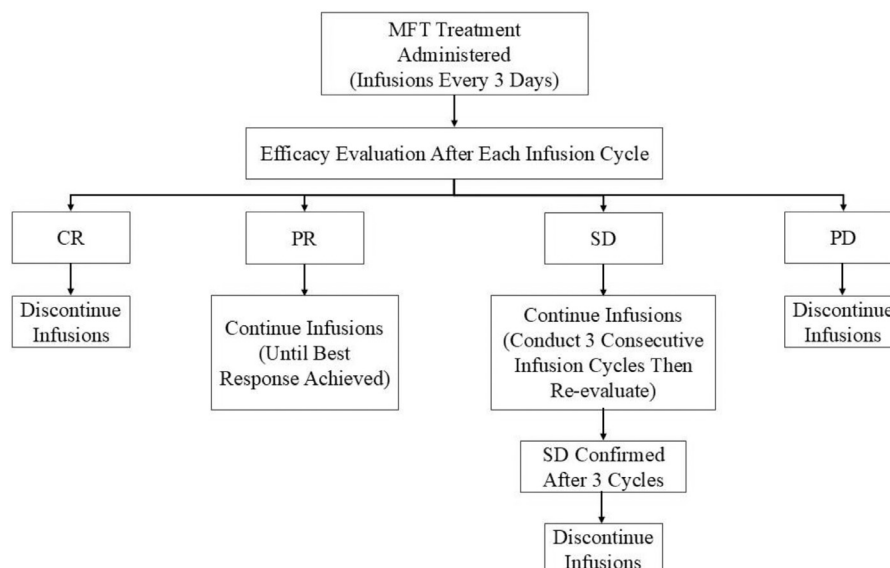


Fig. 1. Decision tree for FMT administration in patients with refractory GI-aGVHD after allo-HSCT. FMT: fecal microbiota transplantation, CR: complete remission, PR: partial remission, SD: stable disease, PD: progressive disease.

death; and progressive disease (PD) was defined as exacerbation of GI-aGVHD (at least one grade higher) with or without improvement of GVHD in other organs; Overall response rate (ORR) was defined as CR plus PR [12,14].

The long-term efficacy evaluation indicators were progression-free survival (PFS) and overall survival (OS). PFS was defined as patients diagnosing refractory GI-aGVHD until GVHD recurrence/progression, death, or the end of follow-up. OS was defined as patients diagnosing refractory GI-aGVHD until death or the end of follow-up. The follow-up period commenced upon the diagnosis of refractory GI-aGVHD.

Safety was evaluated according to adverse events during FMT and follow-up, and adverse reactions were evaluated according to the WHO Adverse Reaction Evaluation Criteria, which were classified as degrees 0 to IV [15].

The deadline for follow-up was September 2023.

Statistical methods

SPSS 25.0 software was applied to statistically analyze the data. For comparisons of baseline characteristics and short-term efficacy between the two groups, normality testing was performed for continuous variables first. Continuous variables conforming to a normal distribution were analyzed using the independent samples Student's *t*-test, while those with a non-normal distribution were compared via the Mann-Whitney *U* test. Categorical variables were evaluated using the chi-squared test, with Fisher's exact test applied when the expected frequency in any cell was less than 5. Fisher's exact test was used to evaluate the factors influencing the recent efficacy. COX proportional hazards analysis was applied to calculate the hazard ratio and 95% confidence interval. Survival curves were drawn by the Kaplan-Meier method. All tests were two-sided, and the difference was considered statistically significant at $P < 0.05$.

Results

General patient characteristics

A total of 51 patients were enrolled, and all had received second-line therapy. Among them, six patients were excluded due to the following comorbidities: one case of ulcerative colitis, one case of thrombotic microangiopathy (TMA), two cases of bacteremia, and two cases of severe pulmonary infection. Based on patient preference,

23 patients received FMT, while 22 patients did not. Three patients were excluded from the statistical analysis due to loss to follow-up. Details are shown in the Figure 2.

The baseline characteristics of the two groups, presented in Table 1, were balanced and comparable with no statistically significant differences ($P > 0.05$). The immunosuppressive drugs administered are summarized in Supplementary Table 1. In the FMT group ($n = 22$), the median age was 26.5 years (range, 7–57 years), with a male-to-female ratio of 15/7. The median daily stool volume was 1300 mL (range, 500–3000mL), the median daily stool frequency was 10 times/day (range, 5–31 times/day) (Table 2). Twenty patients discontinued antibiotics 24 hours prior to FMT, while two patients received aztreonam and vancomycin due to concurrent pulmonary infection. In the control group ($n = 20$), the median age was 28.5 years (range, 9–58 years), with a male-to-female ratio of 12/8. The median daily stool volume was 1350mL (range, 450–2900mL), the median daily stool frequency was 10 times/d (range, 6–33 times/d) (Table 2). In the FMT group, the median frequency of FMT infusions was 3 times/d (range, 1–6 times/d), and individual patient-specific FMT administration details are illustrated in the swimmer plot (Figure 3).

FMT efficacy and safety assessment

The median follow-up time was 88 days (20–2130 days). In the FMT group, the details of FMT administration, remission status, and survival outcomes within 100 days are illustrated in the swimmer plot (Figure 3).

On day 14 (the day after the diagnosis of refractory GI-aGVHD), compared with baseline, the FMT group showed significant decreases in median diarrhea volume (1300 versus 450 mL, $P = 0.017$) and frequency (10 versus 5.5 times/d, $P = 0.049$), whereas the control group demonstrated no statistically significant changes in either median diarrhea volume (1350 versus 900 mL, $P > 0.05$) or frequency (10 versus 9 times/d, $P > 0.05$). Additionally, relative to the control group, both diarrhea volume and frequency were significantly improved in the FMT group on day 14 ($P < 0.05$) (Table 2 and Figure 4).

On day 21, efficacy assessment demonstrated that the rates of CR (31.8% versus 5%; RR = 6.364; 95% CI, 0.856–47.299; $P = 0.047$) and ORR (59.1% versus 25%; RR = 2.364; 95% CI, 1.026–5.447; $P = 0.033$) were significantly higher in the FMT group than in the control group. In the FMT group, two patients (15.3%) experienced GVHD relapse

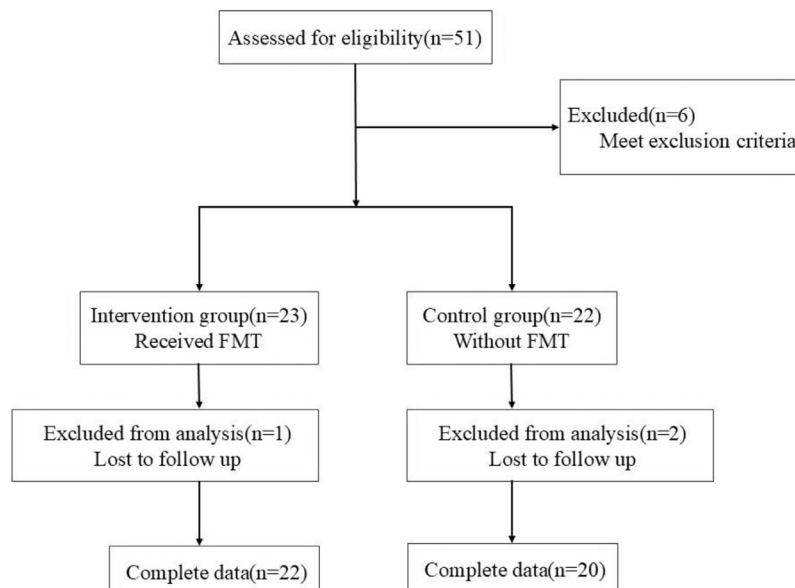


Fig. 2. Experimental flow diagram. FMT: fecal microbiota transplantation.

Table 1
Characteristics of patients with refractory GI-aGVHD after allo-HSCT.

Characteristics	FMT group (n = 22)	Control group (n = 20)	P value
Age [year, M (range)]	26.5 (7–57)	28.5 (9–58)	>0.05
Gender (n)			>0.05
male	15	12	
female	7	8	
Hematologic disease (n)			>0.05
AML	12	11	
ALL	3	3	
SAA	3	4	
MDS	1	2	
PTCL	1	0	
CLL	1	0	
thalassemia	1	0	
Donor type (n)			>0.05
HPD	15	14	
MSD	5	4	
URD	2	2	
Stem cell source			>0.05
PBSC	10	11	
PBSC+BM	2	1	
PBSC+UCB	2	1	
PBSC+BM+UCB	8	7	
Onset organ of GVHD			>0.05
GI	7	7	
GI, liver	4	3	
GI, skin	4	4	
GI, liver, skin	7	6	
Stage of GI-GVHD			>0.05
2	4	3	
3	8	8	
4	10	9	
Median onset time of GI-GVHD (days after HSCT/DLI) [d, M (range)]	30 (15–159)	28 (11–120)	>0.05
Median onset time of FMT (days after GI-aGVHD) [d, M (range)]	21.5 (10–100)	-	-

GI-aGVHD: acute gastrointestinal graft-versus-host disease, allo-HSCT: allogeneic hematopoietic stem cell transplantation, FMT: fecal microbiota transplantation, AML: acute myeloid leukaemia, ALL: acute lymphoblastic leukaemia, SAA: severe aplastic anaemia, MDS: myelodysplastic syndromes, PTCL: peripheral T-cell lymphoma, CLL: chronic lymphocytic leukemia, HPD: haploidentical donors, MSD: matching-sibling donors, URD: unrelated donors, PBSC: peripheral blood stem cells, BM: bone marrow, UCB: umbilical cord blood, DLI: donor lymphocyte infusion.

after achieving ORR, compared with one patient (14.3%) in the control group, and the difference was not statistically significant ($P > 0.05$).

Fisher's exact test showed that among various clinical factors in FMT group, GVHD involvement of organs and bloody stools had a significant impact on the effectiveness of FMT, and the difference was statistically significant. The ORR of GVHD patients with intestinal-only involvement alone and those with simultaneous liver/skin involvement were 100% and 40%, respectively (RR = 2.500; 95% CI, 1.345–4.646; $P = 0.017$). The ORR of patients without bloody stool and those with bloody stool before FMT were 78.6% and 25%, respectively (RR = 3.500; 95% CI, 1.189–10.305; $P = 0.026$) (Table 3).

The main adverse effects of FMT were nausea, vomiting, abdominal pain, and transient low-grade fever, all of which improved with symptomatic management and no serious adverse effects of degree III/IV. One patient developed *Escherichia coli* sepsis 14 days after FMT was performed, which could not be determined to be related to the FMT procedure, and was controlled by anti-infective treatment.

Within 100 days of follow-up, significant differences were observed in both PFS (HR = 2.299; 95% CI, 1.129–4.684; $P = 0.022$) and OS (HR = 2.526; 95% CI, 1.112–5.740; $P = 0.027$) between the two groups (Figure 5). GVHD progression mostly occurs within 40 days and death mostly occurs within 100 days of refractory GI-aGVHD diagnosis. During the follow-up period, both PFS (HR = 2.144; 95% CI, 1.075–4.274; $P = 0.030$) and OS (HR = 2.362; 95% CI, 1.137–4.904; $P = 0.021$) were

Table 2
Evaluation of the recent efficacy in the treatment of refractory GI-aGVHD after allo-HSCT.

Efficacy evaluation	FMT group (n = 22)	Without FMT group (n = 20)	P value
0d			
Stool volume [mL/d, M(rang)]	1300 (500–3000)	1350 (450–2900)	0.962
Stool frequency [mL/d, M(rang)]	10 (5–32)	10 (6–33)	0.695
14d			
Stool volume [mL/d, M(rang)]	450 (100–1600)	900 (150–3000)	0.017*
Stool frequency [mL/d, M(rang)]	5.5 (1–20)	9 (2–28)	0.049*
21d			
CR [n, (%)]	7 (31.8%)	1 (5%)	0.047*
PR [n, (%)]	6 (27.3%)	4 (20%)	0.723
SD [n, (%)]	4 (18.2%)	9 (45%)	0.096
PD [n, (%)]	5 (22.7%)	6 (30%)	0.730
CR+PR [n, (%)]	13 (59.1%)	5 (25%)	0.033*

FMT: fecal microbiota transplantation, GI-aGVHD: acute gastrointestinal graft-versus-host disease, allo-HSCT: allogeneic hematopoietic stem cell transplantation, CR: complete remission, PR: partial remission, SD: stable disease, PD: progressive disease

* Means $P < 0.05$.

significantly higher in the FMT group compared with the control group (Figure 5). The median PFS of the FMT group was 30 (10–2113) days, the median OS was 150 (51–2130) days, the 100-day, 1-year, and 3-year PFS were 40.9%, 28.1%, and 28.1%, respectively, and the 100-day, 1-year, and 3-year OS were 59.1%, 50.0%, and 34.3%, respectively (Figure 5). The median PFS of the control group was 23 (7–1820) days, with a median OS of 80 (20–1901) days. The 100-day, 1-year, and 3-year PFS rates were 5%, 5%, and 5%, respectively, while the 100-day, 1-year, and 3-year OS rates were 20.0%, 10.0%, and 10.0%, respectively (Figure 5). In subgroup analysis, the 100-day, 1-year, and 3-year PFS rates were all 85.7% for patients with GI-only aGVHD, compared with 20%, 20%, and 6.7% for those with concurrent liver or skin involvement (Figure 6). For patients without bloody stools, the 100-day, 1-year, and 3-year PFS rates were 57.1%, 57.1%, and 38.1%, respectively; in contrast, the PFS rates were all 12.5% for patients with bloody stools (Figure 6).

Discussion

GI-aGVHD is mainly due to disturbances in the homeostatic mechanisms of the gastrointestinal immune system and the overreaction of donor cytotoxic lymphocytes [16]. After allo-HSCT, the use of antibiotics and immunosuppressants will affect the diversity and composition of the intestinal barrier and gut microbiota. These changes occur throughout the development of GI-aGVHD, while the dysregulation of the gut microbiota further contributes to secondary inflammatory damage in patients. The study has found that the gut microbiota plays an immunomodulatory role during allogeneic hematopoietic stem cell transplantation [7]. The relative shift in commensal flora towards enterococci after allo-HSCT was particularly pronounced in patients who subsequently developed GVHD, with the proportion of enterococci increasing from 21% to 46% in patients with GVHD, and up to 74% in patients with active GVHD [17]. Pelet et al. also reported a loss of diversity in the gut microbiota during allo-HSCT, with changes strongly associated with aGVHD and prognosis [18]. It can be affirmed that the objective existence of GI-aGVHD combined with severe intestinal dysbiosis suggests the importance of re-establishing the gut microbiota for the treatment of GI-aGVHD.

Currently, the first-line therapeutic agent for aGVHD is glucocorticoids, with an effective rate of 50% for GI-aGVHD, with a survival rate of only 5% to 30% for patients with steroid-refractory GI-aGVHD [3]. In 2016, Fujioka et al. in Japan first applied FMT and its combination for the first time in the treatment of four cases of glucocorticoid-

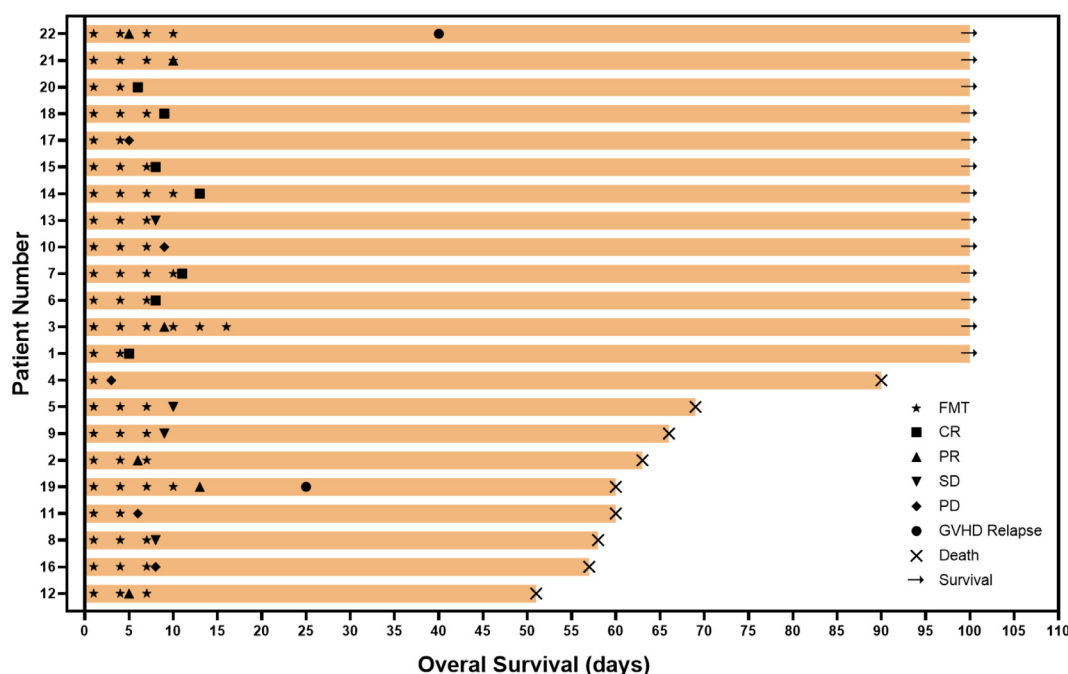


Fig. 3. FMT administration and efficacy assessment in patients with refractory GI-aGVHD after allo-HSCT. FMT: fecal microbiota transplantation, GI-aGVHD: acute gastrointestinal graft-versus-host disease, allo-HSCT: allogeneic hematopoietic stem cell transplantation, CR: complete remission, PR: partial remission, SD: stable disease, PD: progressive disease. (Color version of figure is available online.)

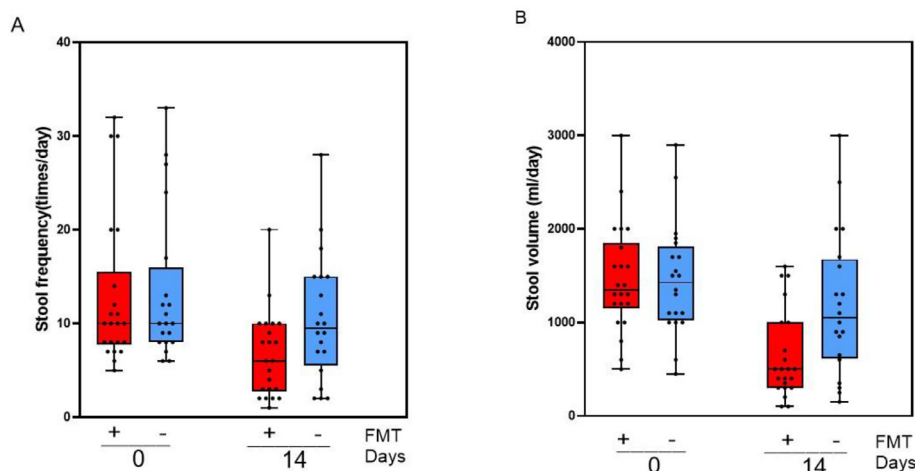


Fig. 4. Clinical response to FMT. (A) Stool volumes of all patients at baseline and Day 14 after refractory GI-GvHD was diagnosed. (B) Stool frequency of all patients at baseline and Day 14 after refractory GI-GvHD was diagnosed. FMT: fecal microbiota transplantation. (Color version of figure is available online.)

Table 3

Relationship between FMT efficacy and general clinical characteristics of patients.

Factor	n	ORR (%)	P value
Gender (male/female)	15/7	66.7/42.8	0.376
Doner type (HPD/non-HPD)	15/7	53.3/71.4	0.648
Onset time of GI-GVHD (≤ 30 d/ > 30 d)	12/10	50/70	0.415
Onset organ of GVHD (intestinal-only/other organ-combined)	7/15	100/40	0.017*
Stage of GI-GVHD (grade 2/ $>$ grade 2)	4/18	100/50	0.115
Bloody stool (Yes/No)	8/14	25/78.6	0.026*
Onset time of FMT (≤ 30 d/ > 30 d)	16/6	62.5/50	0.655

FMT: fecal microbiota transplantation, HPD: haploidentical donors, GI-aGVHD: acute gastrointestinal graft-versus-host disease.

* Means $P < 0.05$.

resistant/dependent intestinal aGVHD [10]. Since then, the use of FMT for the treatment of glucocorticoid-resistant/dependent GI-aGVHD after allo-HSCT has attracted much attention. A prospective study in 2020, which enrolled 15 patients with SR-GI-aGVHD, had an ORR of 66.7% (10/15), and four of the five patients with ineffective treatment died from GVHD complications. FMT Effective treatment was associated with a better prognosis (Mantel-Cox test, $P = 0.0068$) [19]. Another 2020 study included 14 patients with SR-GI-aGVHD, with an ORR of 57% (8/14) and a CR rate of 42% (6/14) for FMT, all non-responders died within 200 days, and those who responded to FMT had a better OS (332d versus 66d; HR: 0.18, 95% CI: 0.03–0.93; $p < 0.005$) [20]. A non-randomized controlled phase I/II clinical study in 2021 enrolled 41 patients with IV glucocorticoid-resistant intestinal aGVHD (23 of whom were treated with FMT), with a CR rate of

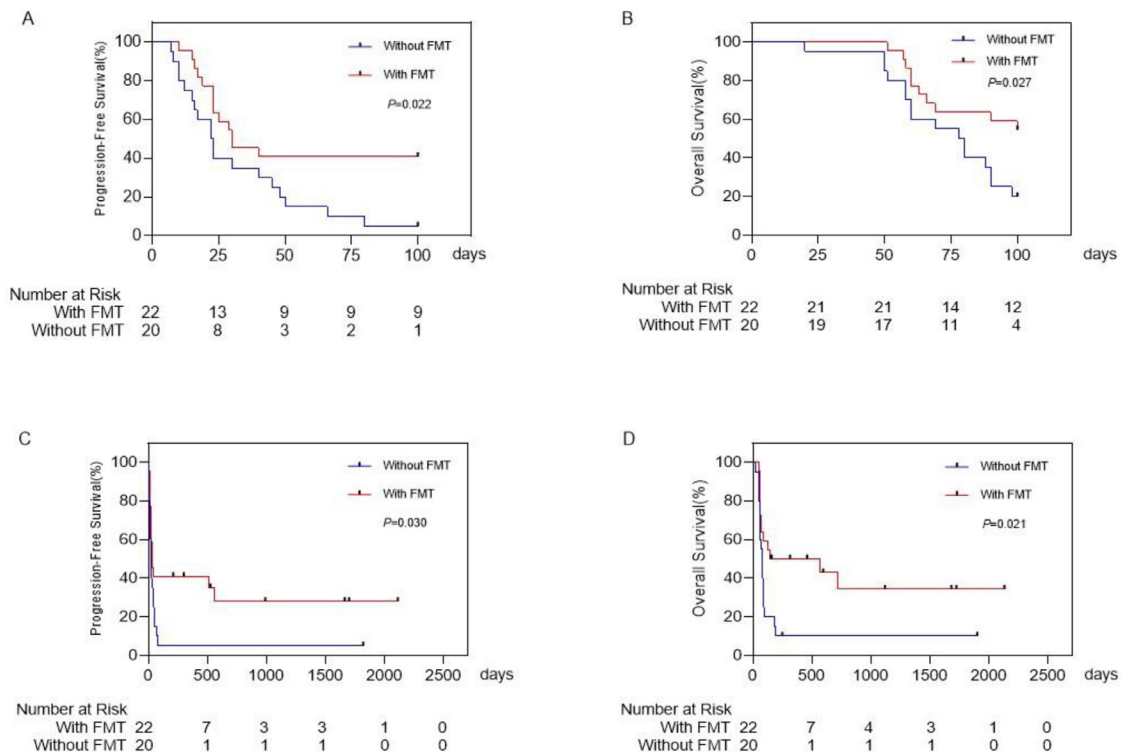


Fig. 5. Kaplan-Meier curves demonstrating survival outcomes. Progression-free survival (PFS) (A) and overall survival (OS) (B) between patients treated with FMT and without FMT in 100 days after the refractory GI-GvHD being diagnosed; PFS (C) and OS (D) at the end of research. Days means day after the refractory GI-GvHD being diagnosed. FMT: fecal microbiota transplantation. (Color version of figure is available online.)

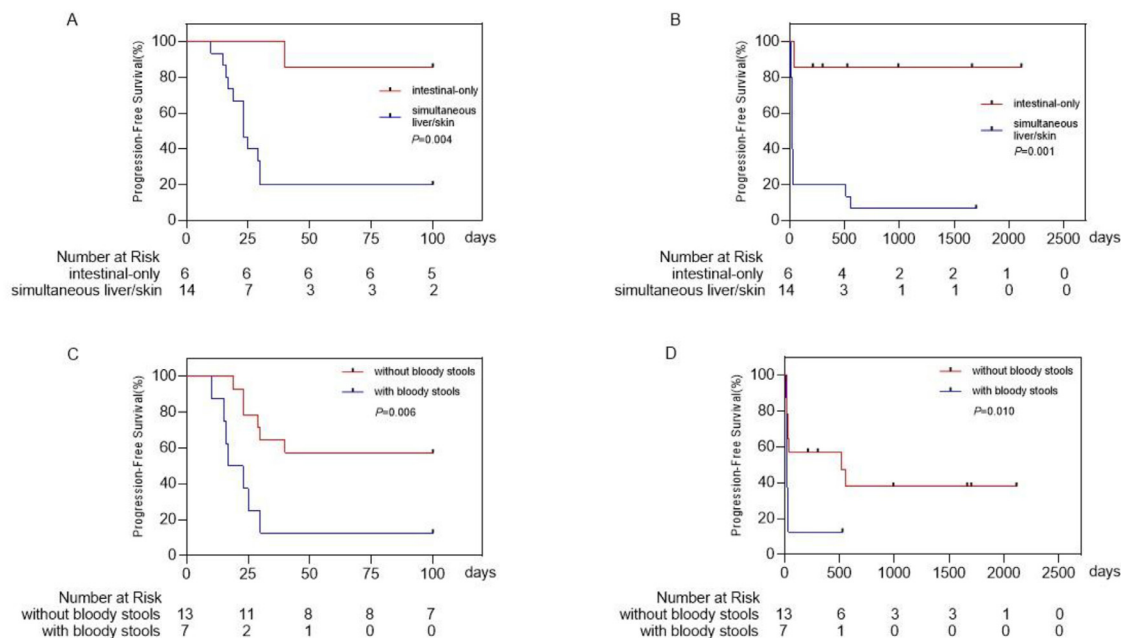


Fig. 6. Kaplan-Meier curves demonstrating survival outcomes in subgroup. (A) and (B) Progression-free survival (PFS) in the intestinal-only and simultaneous liver/skin GI-GvHD patients who treated with FMT were shown. (C) and (D) PFS in the without and with bloody stools patients who treated with FMT were shown. Days means day after the refractory GI-GvHD being diagnosed. FMT: fecal microbiota transplantation. (Color version of figure is available online.)

56.5% (13/23) in the FMT group, which was significantly higher than that of the control group, which was 16% (3/18), and a median survival time at the end of follow-up that was significantly better for the patients in the FMT group than for those in the control group (539d versus 107d, $P = 0.021$) [21].

As GI-aGVHD is characterized by typical extensive intestinal mucosal barrier damage, its safety needs to be taken into account

during the reconstitution of the gut microbiota. A systematic evaluation summarized the global incidence of FMT-associated adverse events (AEs) from 2000 to 2020, with *Clostridium difficile* infection (CDI) as the primary indication, with the main AEs being diarrhea (10%) and abdominal discomfort/pain/cramps (7%), and the incidence of serious adverse events (SAEs), including infections and deaths, was 1.4%, with all SAEs occurring in mucosal barrier-injured patients

(patients with bloody stools) [22]. In the present study, one patient developed *Escherichia coli* sepsis 14 days after FMT was performed, and this patient had the presence of bloody flesh stools suggestive of severe intestinal mucosal damage, although it was not possible to determine whether the infection was related to FMT.

Due to concerns regarding the potential killing effect of antibiotics on the fecal microbiota transplanted via FMT, international guidelines generally recommend discontinuing antibiotics 1–3 days prior to FMT for the treatment of CDIs followed by resuming antibiotic administration after FMT, while antibiotics were not discontinued during the FMT process in some studies [23]. Broad-spectrum antibiotics with anaerobic activity, such as imipenem-cilastatin and piperacillin-tazobactam, can exacerbate the severity of GVHD and increase associated mortality in allo-HSCT recipients by perturbing the gut microbiota (e.g., enriching *Akkermansia muciniphila*), disrupting the colonic mucus layer, and impairing intestinal barrier function; in contrast, antibiotics with weak anaerobic coverage (e.g., aztreonam and cefepime) do not exert such effects [24]. In our study, 20 out of 22 patients discontinued antibiotics 1 day prior to FMT. Antibiotic therapy was initiated based on patients' clinical manifestations—specifically the presence of infectious evidence such as fever and elevated inflammatory markers—following FMT infusion; otherwise, no antibiotics were administered post-FMT.

In conclusion, FMT is effective and safe as a clinical treatment for refractory GI-aGVHD after allo-HSCT, and the efficacy of FMT treatment can be used as a prognostic indicator for the long-term survival of refractory GI-aGVHD. FMT-associated infections should be given special attention.

Author Contributions

Contributions: (I) Conception and design: X Wei and Y Fu; (II) Administrative support: G Li; (III) Collection of data: M Li and F Yuan; (IV) Data analysis and interpretation: M Li and F Yuan; (VI) Manuscript writing: M Li; (VII) Final approval of manuscript: All authors.

Declaration of competing interest

The authors declare that they have no competing interests in this work.

Funding

This research was funded by the Henan Medical Science and Technology Research Project (LHGJ20220187).

Data Availability

The datasets generated during the current study are available from the corresponding author upon reasonable request.

Ethics Approval

The study was approved by the Ethics Committee of the Affiliated Cancer Hospital of Zhengzhou University, China.

Acknowledgments

We are grateful to all our colleagues who provided help. Thank you for unselfishly providing us with clinical work, data analysis, and follow-up.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jcjt.2025.102043.

Article History:

Received 8 May 2024

Accepted 22 December 2025

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